Explanation of the Output

```
lmOut <- lm(medv ~ rm, data=boston_df)
```

```
summary(lmOut)
```

```
##
## Call:
## lm(formula = medv ~ rm, data = boston_df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -23.346  -2.547   0.090   2.986  39.433
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -34.671      2.650  -13.08   <2e-16 ***
## rm             9.102      0.419   21.72   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.616 on 504 degrees of freedom
## Multiple R-squared:  0.4835, Adjusted R-squared:  0.4825
## F-statistic: 471.8 on 1 and 504 DF,  p-value: < 2.2e-16
```

lmOut <- lm(medv ~ rm, data=boston_df)

This means:

- medv = **house price**
- rm = **average number of rooms**

So, the model is trying to answer:

"If a house has more rooms, does the price increase?"

This is a **simple straight-line relationship.** The equation of the line looks like:

$$y = mx + b$$

Where:

- **y** = predicted house price
- **x** = number of rooms
- **m** = slope (how much price changes per room)
- **b** = starting house price when the number of rooms is 0. (intercept)

❖ Call

Call:
lm(formula = medv ~ rm, data = boston_df)

This simply reminds you: You ran a regression predicting **medv using rm**.

Nothing complicated here, it just repeats the model you built.

❖ Residuals

Residuals:
Min -23.346
1Q -2.547
Median 0.090
3Q 2.986
Max 39.433

First understand: **Residual = Prediction Error**

Residual = **Actual price − Predicted price**

So, this section summarizes how wrong the model is.

What do the numbers mean:

Value	Meaning
Min -23.346	Biggest under-prediction. For a house, the model predicted a price that was **23.346 higher than the real price**.
1Q -2.547	For **25% of the houses**, the model predicted the price **about 2.986 units too high**.
Median 0.090	Half of the prediction errors are **above 0.090** and half are **below 0.090**.

3Q 2.986	For **75%** of the houses, the model predicted the price with an error of less than about 2.986 units.
Max 39.433	Biggest over-prediction. For a house, the real price was **39.433 higher than what the model predicted**.

❖ Coefficients (VERY IMPORTANT)

Coefficients:

	Estimate	Std.Error	t value	Pr(>|t|)
(Intercept)	-34.671	2.650	-13.08	<2e-16 ***
rm	9.102	0.419	21.72	<2e-16 ***

Intercept: -34.671 **(The intercept is the value of y when x = 0.)**

Meaning:

- If a house had **0 rooms**, the predicted house price would be **-34.67**.
- This is **not realistic**, but intercepts often don't have practical meaning.

rm coefficient: 9.102

Meaning: Each extra room increases the house price by about 9.1

Std. Error: This tells you **How uncertain the estimate is.**

Example:

rm = 9.102
Std Error = 0.419

This means the true value is around 9.102 but might vary a little.

Small error = more reliable estimate.

t value: This measures how strong the relationship is.

For rm: 21.72

This is **very large**, which means: There is a **very strong relationship between rooms and price**.

❖ Significance Codes

0 '*' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1**

Symbol	Meaning
***	extremely significant
**	very significant
*	significant
.	weak
blank	not significant

Variable rm has: ***

So, it is **very important for predicting prices**.

❖ Multiple R-squared:

Multiple R-squared: 0.4835

This measures how well the model explains the data.

Range:

0 → terrible
1 → perfect

Your value: 0.48

Meaning:

Rooms explain about 48% of house price variation.

So:

- rooms matter
- but **other factors also affect price**

Examples:
- location
- crime rate
- taxes
- schools